



KFTC Announces Results of Cloud Service Market Study

Market tipping and concentration caused by lack of interoperability and burden of data migration

December 28, 2022

Anti-Monopoly Division (Knowledge Industry)

The Korea Fair Trade Commission (led by Chairperson Ki Jeong Han, hereinafter the “KFTC”) announced the results of the cloud service market study.

Cloud computing* is rapidly growing as an essential infrastructure as remote work has become the norm, but there is increasing concern about the lack of market transparency as cloud services require highly specialized technology and expertise by nature. Against this backdrop, the KFTC decided to study the market to understand the overall market situations and competitive conditions. Recently, other overseas competition authorities also conducted such studies: the Japan Fair Trade Commission, French Competition Authority and Netherlands Authority for Consumers and Markets.

* Cloud computing is the on-demand availability of IT resources (e.g. servers, storage and networks) without direct ownership and management by a company. Most cloud computing services generally fall into three categories: infrastructure as a service (IaaS), platform as a service (PaaS) and software as a service (SaaS).

The researchers from inside and outside the KFTC conducted two different surveys in a consecutive way and comprehensively analyzed the survey findings: (1) the first survey of 32 major cloud service providers; and (2) the second survey of about 3,000 stakeholder companies: (i) clients, (ii) managed service providers (MSP) and (iii) solution providers that release software via cloud marketplaces operated by cloud service providers.

In Survey 1, 32 major cloud service providers were requested to submit detailed information in written format* regarding the current market situations and business practices. In Survey 2, stakeholder companies in a business relationship with the cloud service providers were requested to complete a questionnaire** to comprehensively analyze the service contract terms and competitive

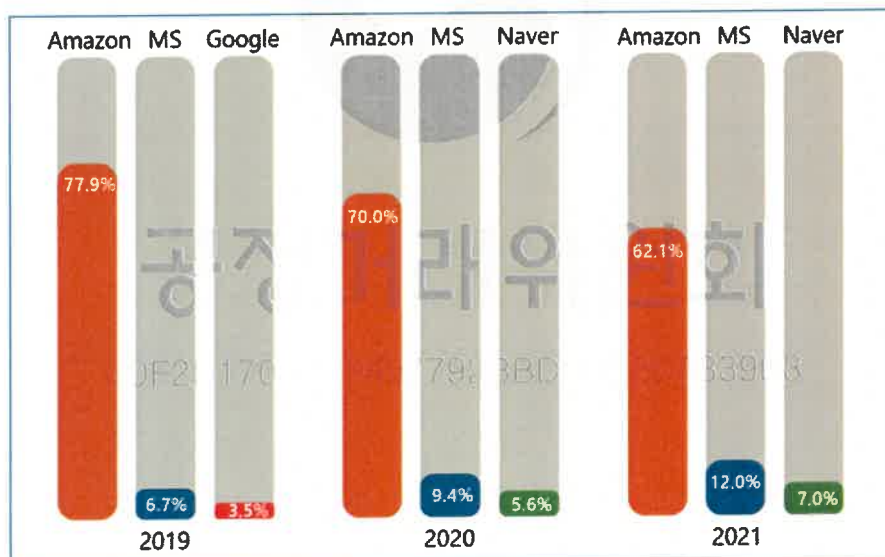
conditions.

* Questionnaire items: products and revenues, market situations, business relationships (e.g. clients, distribution partners, marketplace solution providers, associated software providers), tip-off and opinions, etc.

** Questionnaire items: structural factors that hurt fair trade, experience of unfair practices, demand-side substitutability for market definition, etc.

According to the survey, the domestic cloud market is highly concentrated with a few global big techs. Over the recent three years from 2019 to 2021, Amazon Web Services (AWS) ranked first, accounting for 62-78%, followed by Microsoft (MS) at 6-12% in terms of market share*. Google and Naver, which ranked third and fourth respectively, have been expanding their market share.

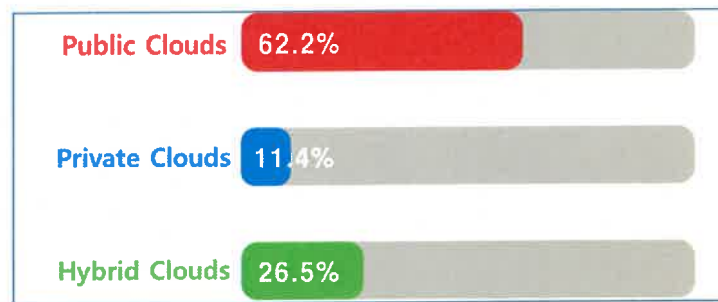
< Cloud Market Share of Three Major Companies in Korea (2019-2021) >



* The market share was calculated based on IaaS and PaaS revenue data from 16 cloud service providers whose services focus on public cloud among the respondents surveyed.

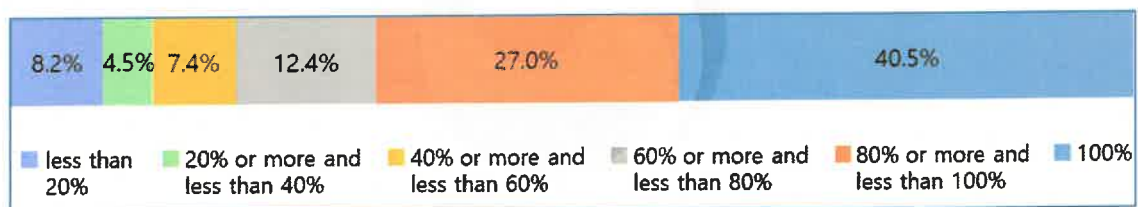
In terms of types of cloud computing, 62.2% clients used public clouds, 26.5% clients used hybrid clouds (a combination of public and private cloud) and 11.4% clients used private clouds.

< Cloud Environments in Use >

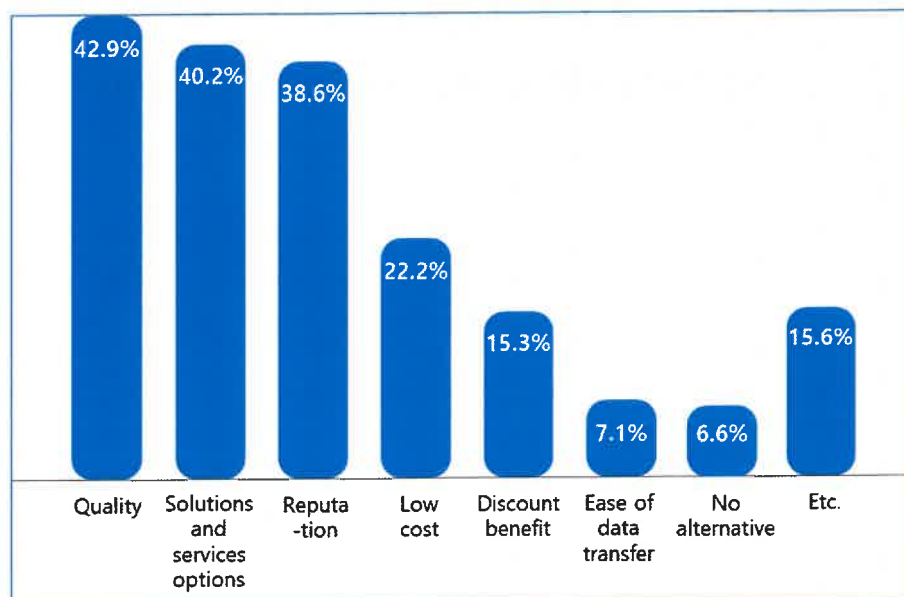


Clients were likely to be dependent on a single service provider rather than using multiple providers as 79.9% of client firms responded that they pay more than 60% to their primary provider of the total cost spent on cloud services. Product and service differentiation was the important factor in their choice as the following were the major reasons of such high dependence: quality (42.9%), solutions and services options (40.2%) and reputation (38.6%).

< Percentage of Primary Provider Out of Total Cloud Spending >

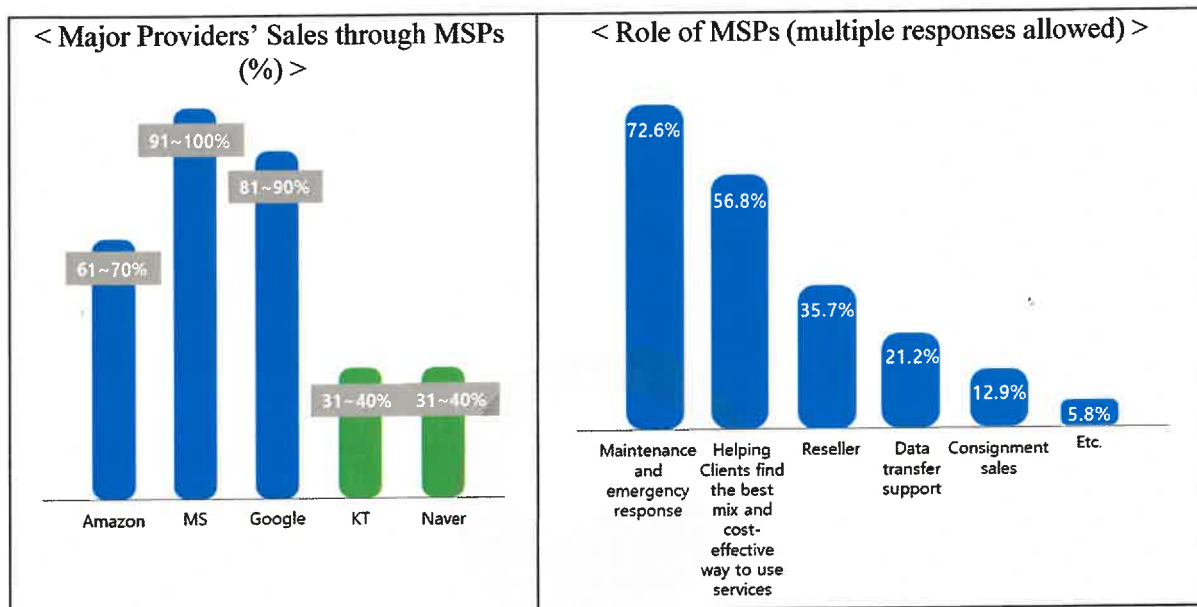


< Reason for choosing their Primary Provider (multiple responses allowed) >



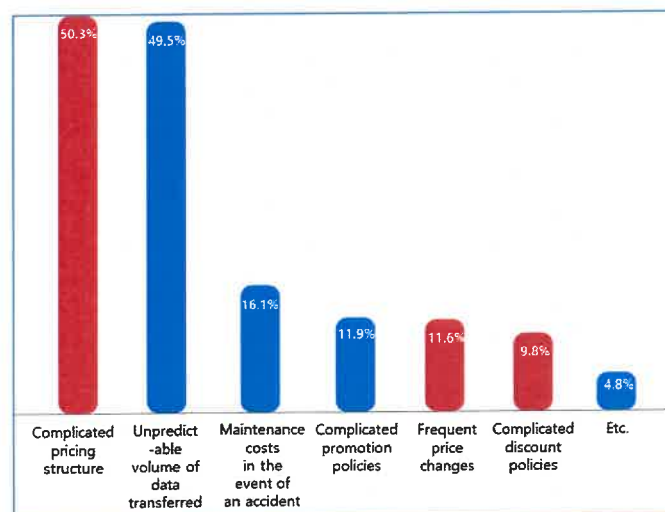
When it comes to the delivery of cloud computing, global big techs, such as Amazon, MS and Google,

preferred partnering with MSPs instead of directly dealing with their clients. Most of the MSPs partnered with multiple cloud service providers rather than exclusively working with a single provider. According to the responses of clients that use cloud services delivered by MSPs, MSPs are mainly responsible for providing maintenance and emergency response (72.6%) and helping clients find the best mix and cost-effective way to use services (56.8%).

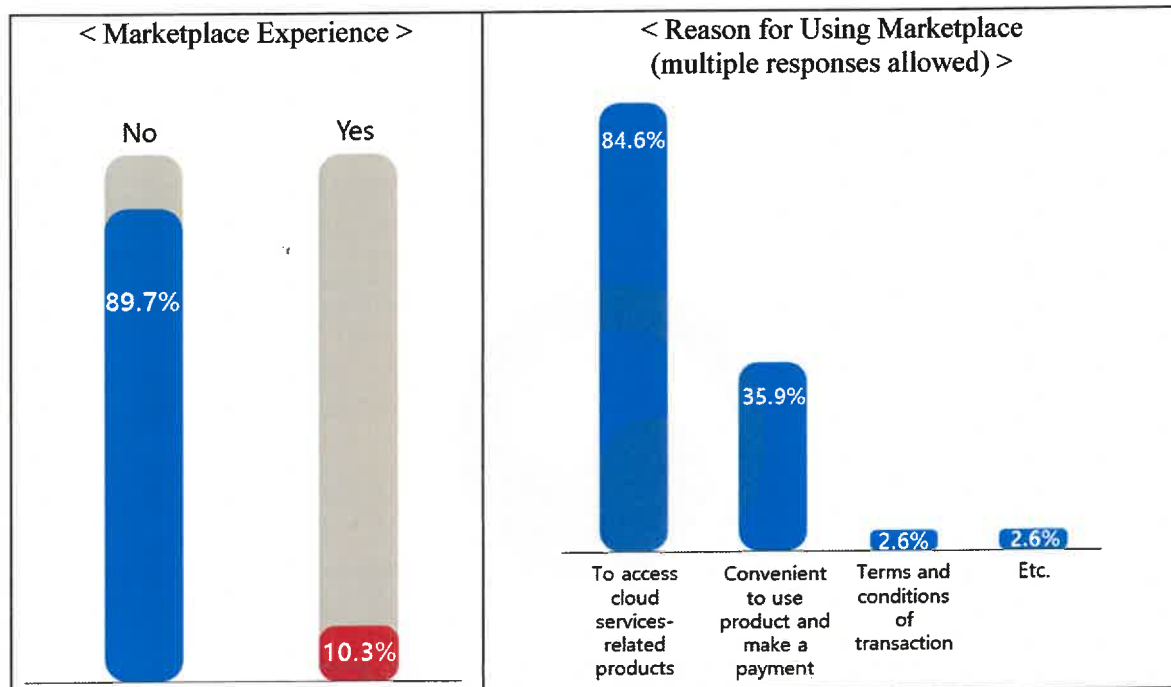


The survey also showed that only 20.1% of clients found it easy to predict the cost of cloud services. Others cited complicated pricing structure (50.3%) and unpredictable volume of data transferred (49.5%) as the reason for low cost predictability.

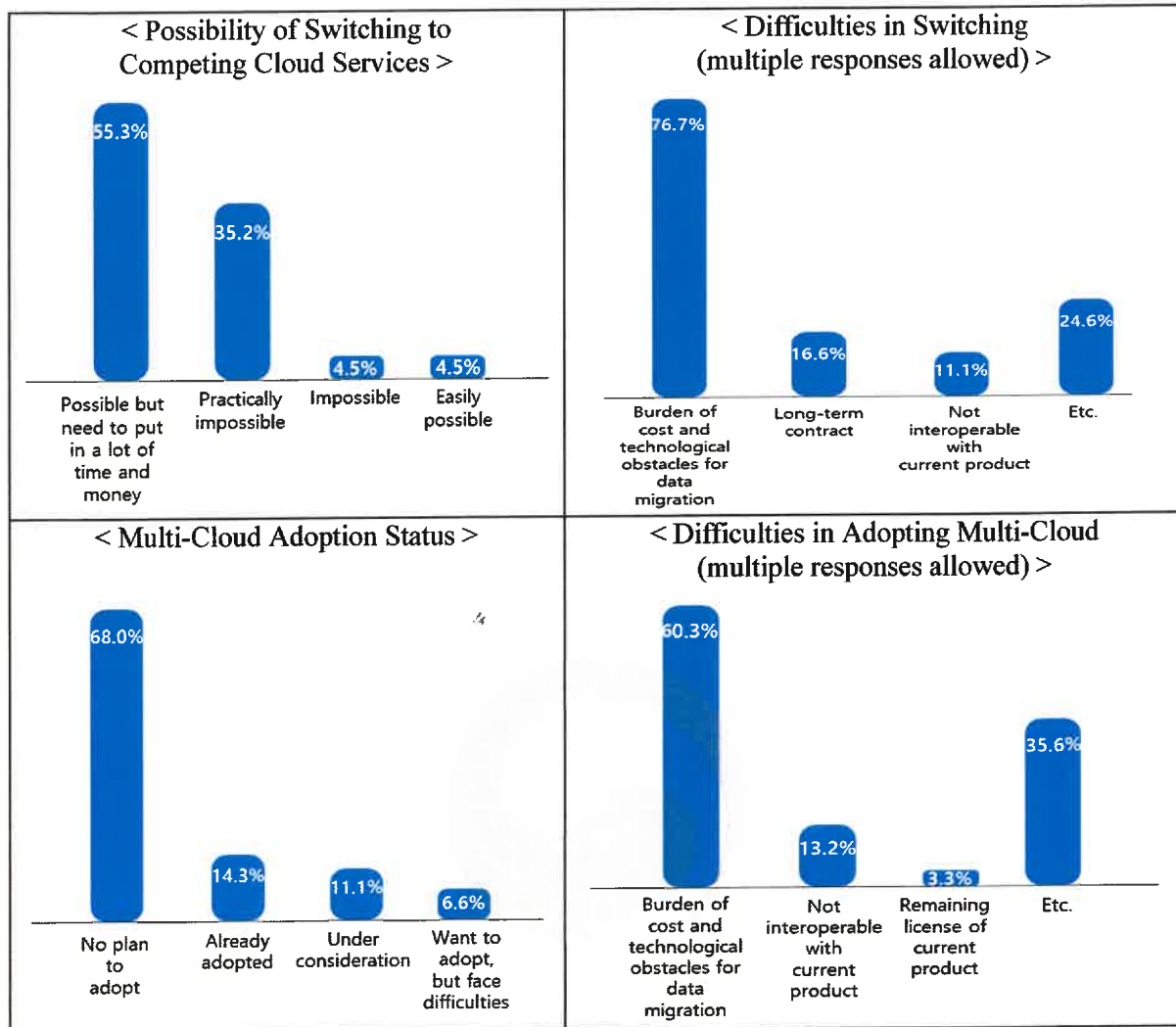
< Reason for Low Cost Predictability (multiple responses allowed) >



Most of the major cloud service providers—including Amazon, MS, Google, Naver and KT—operate a solution marketplace that acts as an intermediary between solution providers and clients. Solution providers can offer various paid or free software and services to clients in cloud marketplaces, and the intermediary charges 3-20% commission fee on transactions. However, in Korea, a mere 10.3% of domestic clients said that they have ever used a cloud marketplace.



When it comes to anticompetitive factors in the cloud market, the research found various constraints clients face regarding interoperability and data transfer in switching to competing cloud services or using multiple services (multi-homing). Interoperability is not guaranteed when companies migrate to rival cloud services or adopt a multi-cloud strategy, which means they have to redesign or redevelop their existing work environment—e.g. programming languages, application programming interfaces (API), software development kits (SDK). When switching to competing cloud services, they face data transfer challenges across different cloud environments, which takes a lot of time and money. As a way of acquiring customers, cloud service providers usually do not charge for inbound data traffic but charge for outbound data traffic in a data migration.



The research indicated that there is a need to pay close attention to anticompetitive business practices in the market: whether service providers harm competition through bundling or self-preferencing and whether they use unfair terms in consumer contracts. Yet, regulators should comprehensively review the circumstances, such as how competition plays out and what disadvantages clients suffer in the domestic market.

Cloud computing is a key platform across all industries, so establishing an even playing field is important. It turned out, however, that the domestic cloud market is highly concentrated with a few global big techs and that the market competition tends to be restricted due to lack of interoperability and data concentration. Based on such research results, the KFTC will set the overall approach to antitrust policies in the cloud market that can effectively improve relevant systems and increase competitive pressure while keeping an eye on any potential risks that could harm competition or put domestic firms in an unfavorable position.

**The Korean text of the documents is confirmed to be authentic and English version is only for reference*

4



대한민국